



CD-A twin niches in the Mont Terri Rock Laboratory: Characterization and interpretation of hydraulic parameters with regard to safety aspects

Gesa Ziefle^{a,*}, Tuanny Cajuhi^a, Stephan Costabel^a, Antoine Fourrière^a, Markus Furche^a, Jana Gerowski^a, Tilo Kneuker^a, Franz Königer^b, Jobst Maßmann^a

^a Federal Institute for Geosciences and Natural Resources (BGR), Hanover, Germany

^b Ingenieurgesellschaft für Sensorik in der Umwelttechnik mbH (ISU), Karlsruhe, Germany

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ABSTRACT

In the Mont Terri Rock Laboratory (Switzerland), a comprehensive interdisciplinary exploration program is carried out to increase knowledge about long-term coupled hydraulic-mechanical effects in Opalinus Clay. As these effects are of significant interest regarding the stability and integrity of a potential storage facility for high radioactive waste, the “Influence of humidity on cyclic and longterm deformations” (CD-A) experiment aims to characterize the Opalinus Clay, especially its heterogeneous sandy facies, and to investigate the evolution of the water content, pressure and deformation in the formation. This article provides an overview of measurement examinations and geologic investigations carried out in the context of the CD-A experiment. The twin niches of the CD-A experiment are located in the upper sandy facies of the Opalinus Clay which is characterized by the existence of different subfacies types and fault zones. This contribution focuses on the detailed characterization of the claystone, while the evolution of the water content. Combined geophysical and geological investigations during the first two years after the excavation of the niches have been carried out. They indicate that the subfacies type and pre-existing fault zones do not affect the distribution of water content severely, which could be an essential finding in the context of detection procedures for potential high level waste (HLW) repositories. Furthermore, miniseismic as well as pulse tests indicate clearly, that the extent of the excavation damaged zone (EDZ) strongly depends on the climatic conditions in the underground galleries: After 1.5 years the EDZ and the increase of permeability is much more pronounced in the open than in the closed twin. Future work aims on an increased process understanding based on a verified constitutive model for Opalinus Clay and a comprehensive system understanding considering long-term effects due to climatic conditions as well as the impact of geotechnical boundary conditions and heterogeneities.

1. Introduction

The comprehension of coupled hydraulic-mechanical (HM) effects in claystone is of significant importance for the safety assessment of a potential repository for high-level radioactive waste (HLW). In particular the various coupling effects are to be regarded as relevant: For example, the water content not only influences numerous material parameters but also potential plastic effects and crack formation behavior. These mechanical effects in turn have a direct influence on stress state and potential flow processes and thus on the integrity and also stability of the repository. Consequently, the applicability of constitutive material models for claystone and thus also the safety assessment is essentially dependent on the correct formulation of the hydraulic-mechanical couplings. Here, the constitutive material model and also the correct parametrization of the material are of significant

importance. As described, the coupled effects depend essentially on the hydraulic situation in the Opalinus Clay - especially on the water content, which is not necessary homogeneous. Therefore, a focus of the presented investigations is also related on material heterogeneities and their impact on safety aspects. The multi-disciplinary investigations and their results form a basis for a well-founded characterization of the rock, taking the water content into account. The presented investigations are carried out in the context of the “Influence of humidity on cyclic and longterm deformations” (CD-A) experiment as part of the Mont Terri Project, which is introduced in Section 2. This work is motivated by the need of a detailed understanding of the geotechnical system and its properties as it is required to evaluate the stability and integrity over long time periods. It provides a comprehensive characterization and combined interpretation of the material based

* Corresponding author.

E-mail address: gesa.ziefle@bgr.de (G. Ziefle).

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on geologic and geophysical investigations, building the basis for numerical approaches to capture coupled hydraulic-mechanical effects. Claystone is a very complex material, characterized by anisotropy and often heterogeneities. These have significant impact on the hydraulic and mechanical boundary conditions, like the moisture content and the stress state, on the material behavior. Various laboratory, in-situ and theoretical investigations with this regard have also been carried out and discussed in the literature (Refs. 1–4 amongst others), indicating a high impact of the claystone composition on the material parameters and especially on the transport properties like gas and water permeabilities and on the swelling behavior. Besides the anisotropic conditions, the saturation plays an important role on the coupled effects, influencing the stiffness, strength and swelling capacity of the rock, as discussed in Refs. 3, 5, 6. Another specific focus of various investigations is the interplay between the material properties, e.g. permeability, and the evolution of the EDZ.^{7–9} In this context, a stronger influence of coupled HM effects is observed in fractured regions, e.g. increase of permeability. Ref. 10 states that the evolution of permeability in excavation damaged zones is crucial to investigate, especially during cyclic loading. Due to the importance of this kind of material conditions, geophysical and geohydraulic methods are highly requested in claystone research. In various geophysical fields of interest, research aims on a relationship between geophysical logging data and mechanical material properties. In the last years, various constitutive approaches for effects in claystone are discussed and illustrate the difficulty of correctly representing the complex material behavior even under consideration of various boundary conditions. Ref. 11 presented a pragmatic approach to abstract the excavation damaged zone around tunnels of a geological radioactive waste repository in numerical models. Combined presentations of experimental and numerical investigations are presented in Refs. 12, 13. For handling coupled HM effects in numerical simulations, coupled numerical approaches have been proposed in the ninetens by Refs. 14, 15 amongst others. The work on hand focuses on the geologic interpretation and the description of various types of geophysical measurements conducted in boreholes around an excavation aiming on a characterization of the host rock including heterogeneities at different scales such as subfacies types, fault zones and the zone, which is affected by the excavation. In this context, the literature distinguishes between excavation damaged zone (EDZ), with irreversible changes and the excavation disturbed zone (EdZ), where changes may be in the elastic range coming along with reversibility. Here, the designation EDZ (excavation damaged zone) is used, even if the geophysical measurements do not necessarily provide information about the reversibility. The combined interpretation - also focusing on the water content distribution - provides a detailed characterization of the host rock and additional information about hydraulic parameters, like resistivity, seismic wave velocity, lithologic description, permittivity, permeability, and water content. The setup of the CD-A experiment and its twin niches enable to compare these parameters under different conditions. For that, we discuss the corresponding results directly and within 1.5 years after the excavations and provide an overview of the strengths and limitations of the different investigation strategies. The impact of the opening conditions as well as heterogeneities is discussed and a comprehensive characterization of the Opalinus Clay with additional information on hydraulic parameters can be provided. The presented data aims on further understanding of hydraulic effects and related further development of coupled constitutive material models in heterogeneous claystone.

2. CD-A experiment in the Mont Terri project

The presented investigations are carried out in the context of the “Influence of humidity on cyclic and longterm deformations” (CD-A) experiment as part of the Mont Terri Project, which is introduced in Ref. 16. The Mont Terri Project is an international research project aiming at the geological and geotechnical characterization of Opalinus

Clay as summarized in Ref. 17. The Mont Terri rock laboratory is located in the western part of Switzerland in the Jura mountains and hosts various field experiments to investigate the coupled thermal, hydraulic, mechanical, chemical and biological (THMBC) processes in the claystone. Information about the lithology of the Opalinus Clay in the area of the Mont Terri rock laboratory can be found in Refs. 17–19. The tectonic structure of the main fault is discussed in Ref. 20, and the geomechanical behavior at multiple scales is summarized in Ref. 21. Ref. 22 discuss multidisciplinary data from petrographic and structural studies, geophysical borehole logging and permeability tests to assess the impact of brittle faults on barrier capability. And an updated review of the rock mechanics properties of the Opalinus Clay of the Mont Terri rock laboratory based on laboratory^{23–25} and combined with field testing can be found in Ref. 26. The aim of the CD-A experiment is the evaluation of HM long-term processes. The experiment is characterized by a comprehensive measuring program including geological, geotechnical and geophysical characterization. The unique setting of the twin niches enables an understanding of the unrevealed influence of varying climatic conditions on the host rock. It assesses the impact of the lithology on the subfacies scale on barrier relevant properties like water content and resistivity. Furthermore, it investigates the extent and hydraulic characteristics of the EDZ with regard to potential flow paths. This article summarizes the results of the first two years after excavation of the twin niches.

The excavation of the CD-A twin niches has been carried out as part of the extension of the Mont Terri Rock Laboratory (Ref. 27) in September 2019. The niches were excavated in fall 2019 near the Gallery 2018 in an area characterized by the upper sandy facies of the Opalinus Clay as shown in Fig. 1. With a diameter of 2.3 m and a length of 11 m, both are oriented perpendicular to the strike of bedding. The surface of the claystone in the niches is not covered with any kind of stabilizing support. Consequently, a direct interaction between the humidity and temperature in the niche and the claystone is possible and measurable. The aim of investigating the twin niches is a direct comparison of the different climatic conditions within the niches (open/closed) and their impact on the hydraulic-mechanical effects in the host rock. While the open twin is characterized by the seasonal variation of temperature and relative humidity as in the rest of the rock laboratory, the conditions in the closed twin provide a constantly high relative humidity and very low changes in the temperature as well. As a result of these boundary conditions, there will be an ongoing desaturation of the host rock around the open twin, which is not expected around the closed twin. The evolution of relative humidity as measured in the niches is depicted in Fig. 2.

The complex measuring program of the experiment focuses on the climatic conditions within the twin niches, the characterization of the host rock, and the evolution of the deformation, saturation and pore pressure as it is presented in Ref. 28 and Ref. 29 and summarized in Fig. 3 and Table 1. Within a few weeks after excavation, the measuring system (Extensometer, Taupe system, Mini-Piezometer, Climatic devices, Convergence measurements and further more) in and around the niches has been installed and the climatization has been adjusted as described in Ref. 16. Beside the boreholes that are used for the permanent installation of measurements like TDR technique-based Taupe system, mini-piezometer and extensometer measurements, several additional boreholes have been drilled in September 2019 and March 2021 to investigate the Opalinus Clay around the niches with core and borehole measurements. Methods like the electric resistivity tomography (ERT), the nuclear magnetic resonance measurements (NMR), the Calcium Carbide method (CCM), pulse tests, evapometer measurements, borehole and core scans are used in the context of characterization of the rock. They provide information about certain material parameters like the electric resistivity, the permittivity and process variables like the water content directly after drilling. Furthermore, geological investigations of drill cores and of the rocks cropping out at the surface of the niches and the gallery walls have been carried out (Ref. 30).

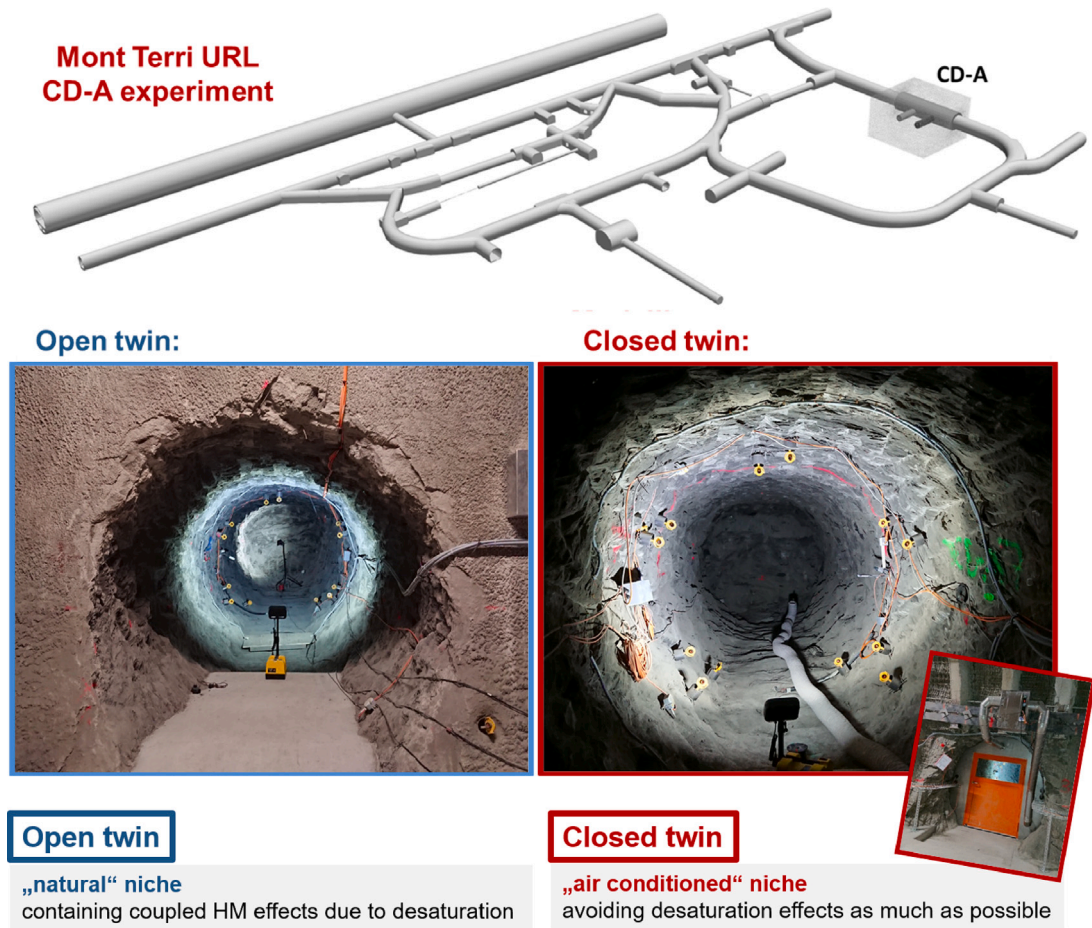


Fig. 1. Top: Location of the CD-A experiment in the Mont Terri URL (top) and view into the open (bottom left/blue frame) and the closed (bottom right/red frame) niche. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

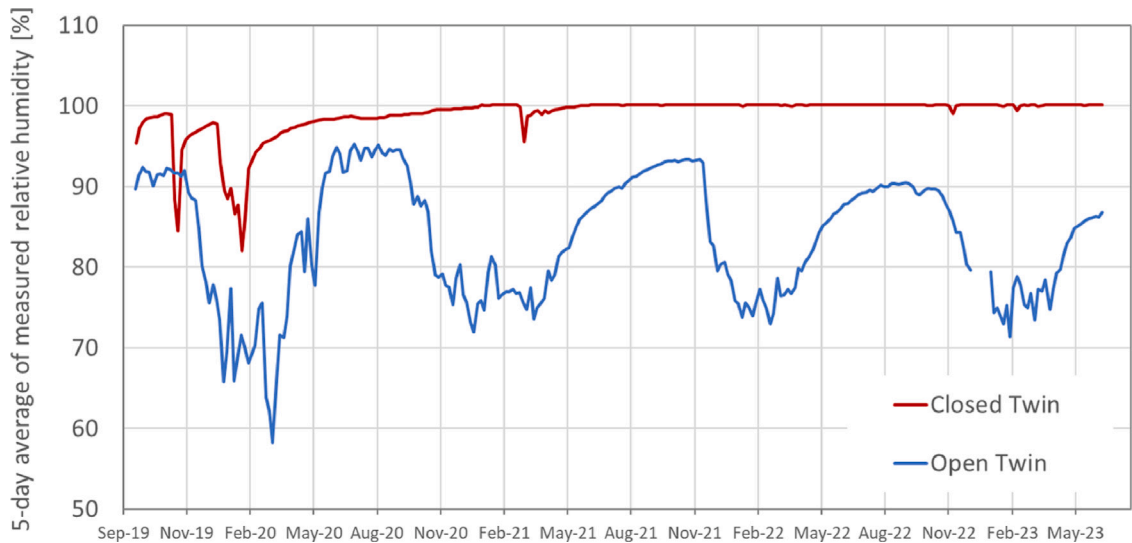


Fig. 2. 5-day average of measured relative humidity within the twin niches.

A comparative interpretation of the investigated parameters provide information concerning the distribution of different litho-stratigraphic (sub)units, the extent of an EDZ and a borehole disturbed zone (BDZ) as well as the occurrence and properties of tectonic fault zones and

individual faults (Table 2). To complement the experimental campaign, numerical calculations are performed to provide information about the water content and the importance of the mentioned heterogeneities on the coupled HM processes.³¹ Additional measurements focusing on

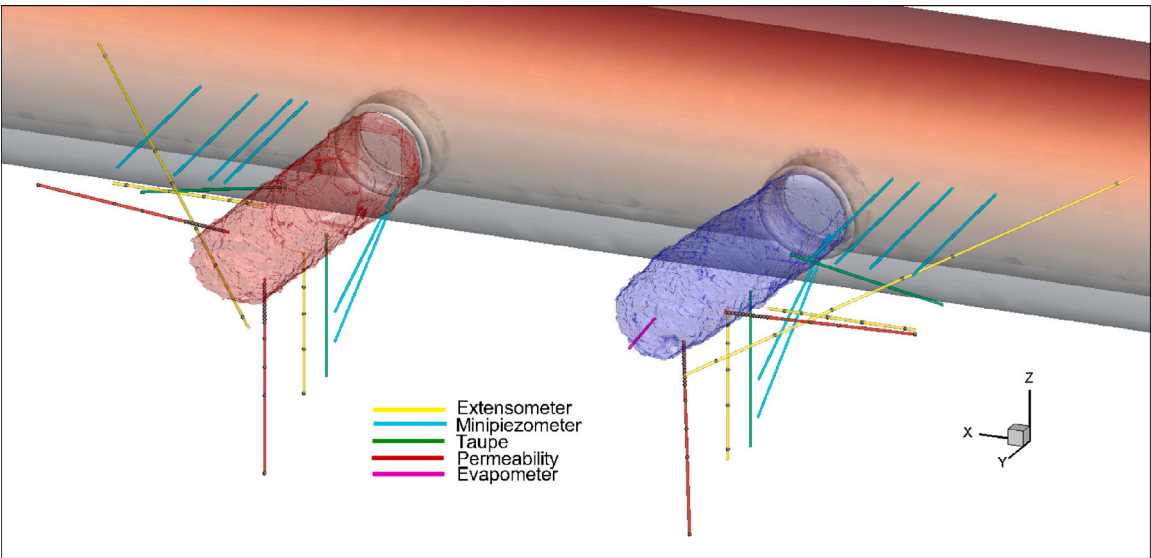


Fig. 3. Location of boreholes in the vicinity of the CD-A twin niches.

Table 1
Measurement campaign of the CD-A experiment. The investigations discussed here focus on the characterization of claystone and the related saturation including the information about the climatic conditions.

To be investigated	Kind of measurement	Parameter	Unit
Characterization of claystone	ERT (borehole)	Electric resistivity	$\Omega\text{ m}$
	ERT (niche profile)	Electric resistivity	$\Omega\text{ m}$
	MSM (borehole)	Elastic wave velocity	m/s
	permeability (borehole)	permeability	m^2
	Geologic characterization	Lithology	–
Climatic conditions		Air humidity	%
		Temperature	$^{\circ}\text{C}$
		Atmospheric pressure	Pa
Saturation	Taupe measurements	Permittivity	–
	NMR (core)	Water content	m^3/m^3
	CCM (core)	Water content	m^3/m^3
	NMR (niche surface)	Water content	m^3/m^3
	Evapometer	Conductivity	m/s
Pressure	Mini-piezometer	Pore pressure	Pa
	Psychrometer	Suction	Pa
	Thermo-hygrometer	Suction	Pa
Deformation	Extensometer	Displacement	m
	Convergence (niche)	Displacement	m
	Laserscan (niche profile)	Displacement	μm

the deformation behavior such as laser scans, convergence and extensometer measurements are not part of this article and are discussed e.g. in Ref. 32.

The characterization of the Opalinus Clay around the twin niches is carried out with geologic and mineralogical investigations as well as electric resistivity measurements (ERT), use of the mini seismic method (MSM), pulse tests and evapometer measurements. Focusing on the water content around the niches, the nuclear magnetic resonance method (NMR), the Calcium-Carbide Method (CCM) and Taupe/TDR measurements are used.

3. Geology and lithology

This paper focuses on a combined interpretation of lithological investigations with related geophysical measurements, also aiming on the determination of the water content. Similar approaches for the characterization have already been discussed in Ref. 33, who conducted multidisciplinary investigations of Mont Terri drill cores to capture the structural and compositional variability of different sub-facies types

of Opalinus Clay. The study states; the variability depends significantly on the direction relative to the bedding plane and underline the proposed composition scheme of geological subfacies types conducted by Ref. 34. Furthermore, the presented investigations point out the impact of structural and compositional heterogeneities on barrier properties like mechanical behavior, radionuclide retention capacity, swelling capacity and also the behavior under thermal load. Further in-situ investigations, as reported in Refs. 33–35, use high-resolution seismic tomography for the characterization of the anisotropic material behavior of the shaly and carbonate-rich sandy facies of the Opalinus Clay in the Mont Terri rock laboratory. A seismotectonic analysis of the Mont Terri rock laboratory has been presented by Ref. 36. In general, the Opalinus Clay in the Mont Terri Rock Laboratory can be divided into three facies types: the shaly, the sandy and the carbonate-rich sandy facies type. Until recently, the shaly and the lower sandy facies type of the Opalinus Clay were in the focus of the geological investigations.³⁷ The construction of the new Gallery 2018, where the twin niches of the CD-A experiment are located, now provides the opportunity for a detailed characterization of the upper sandy facies type. A detailed study and interpretation of the geology around the twin niches can be found in Ref. 30.

The sandy facies consists of a silty-sandy claystone with abundant thin elongated sand lenses, mm-thick bedded or laminated sand layers, calcareous and often bioturbated micritic nodules and bioterritic horizons. Quartz grains inside the sand layers are commonly cemented by carbonates, however, the cementation is not always continuous. In the Mont Terri Rock Laboratory, the sandy facies can be subdivided into an upper and a lower sandy facies type. Both are characterized by differences in composition. Compared to the lower sandy facies, the upper sandy facies includes a higher quantity of clay-rich interlayers. Furthermore, the niches are characterized by variously dipping faults and fault zones that are summarized in Fig. 4. The geologic investigation indicates fault planes parallel to the bedding as well as non-bedding parallel fault planes are present. A disintegration of the fault rock along brittle micros shears into duplex-like tectonites was observed at several locations. This scaly clay fabric indicates distributed shear along numerous discrete fault planes (compare to Ref. 38). Some of the encountered fault planes are filled with calcite (Ref. 30).

3.1. Core and borehole investigations

High-resolution core photographs were taken using a DMT CoreScan3 system specifically designed to take 360° photographs. For this

Table 2
Investigation methods and available information.

Method:	Lith.	ERT	MSM	NMR/CCM	Taupe	Pulse test	Numerics
Info:							
Subfacies type	+	+	–	–	–	–	(*)
EDZ	–	–	+	–	+	(+)	(*)
Fault zones	+	–	–	–	(+)	(+)	(*)
Water content	–	+	–	+	(+)	–	+

[+: information available, -: no information available. *: may provide information about their importance on physical effects, (): limited information]. Lith. = Lithological investigations on drill cores.

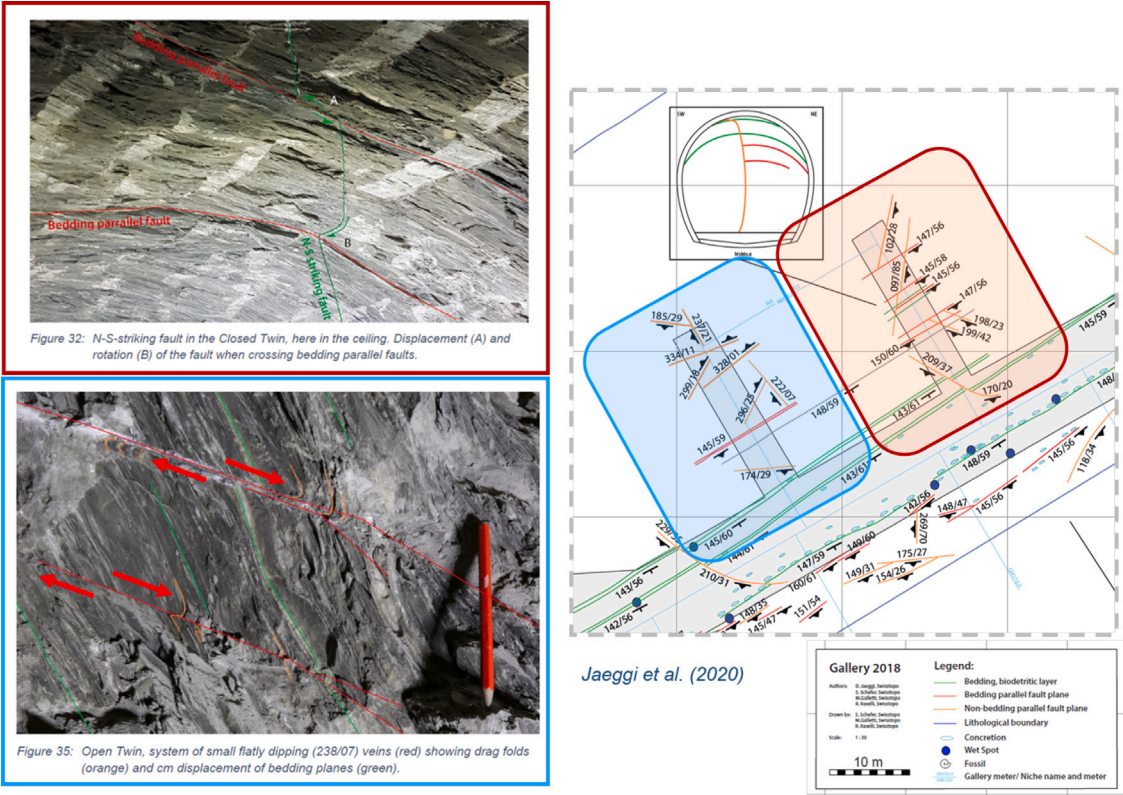


Fig. 4. Structural geology of the twin niches.
Source: Extracted from Ref. 30.

purpose, the core is rotated around its longitudinal axis while a digital camera records the picture. Core scans have been carried out on drill cores from all studied boreholes (BCD-A1 to BCD-A4, BCD-A7, BCD-A8, BCD-A24 to BCD-A27) while optical borehole images (OBI) have been acquired using an QL40-OBI-2G borehole televiewer (Mount Sopris Instruments) in boreholes BCD-A24 to BCD-A27. Borehole scanner data was captured by a Mount Sopris SCOUT data acquisition system. Furthermore, water content investigations at the cores have been carried out using NMR (nuclear magnetic resonance) and CCM (Calcium-Carbide) methods as will be explained in Sections 4.5 and 4.6. A summary of these investigations can also be found in the supplementary data.

3.2. Lithological interpretations

Structural–lithological investigations form the basis for rock characterization and exploration. On site, the swisstopo team first described the non-oriented drill cores visually. The drill core material from the CD-A experiment niches has been further characterized at the BGR core repositories in Berlin and Hannover (Germany).

The investigated drill cores capture one of the five major facies types of the Opalinus Clay and therefore allow to study the intra-facies

variability of the upper sandy facies type. In this context, the subfacies classification proposed by Ref. 34 provides a framework for an in-depth characterization of lithological intra-facies variations for the Opalinus Clay. Each subfacies type is characterized by a distinctive color, grain size, fabric and lithology. By combining the high-resolution core scans with detailed petrographic-structural observations directly on the drill core material, lithological profiles including the subfacies types classification according to Ref. 34 were produced and are presented in Fig. 10, Fig. 11, Fig. 8, and Fig. 9 in comparison with the results of other investigation techniques. In order to further quantify the intra-facies variability, it is planned to perform additional microstructural and mineralogical–geochemical analyses on the core material in the near future.

4. Measurement examination

In this section the various measurement methods are introduced: Electric Resistivity Tomography (ERT), Mini Seismic Method (MSM), the pulse tests, evapometer measurements, Time Domain Reflectometry (TDR/Taupe), Nuclear Magnetic Resonance (NMR), and Calcium Carbide Method (CCM). These measurements focus on the characterization of the claystone by investigating the electric resistivity, seismic velocities, water content, the and the permeability of the host rock.

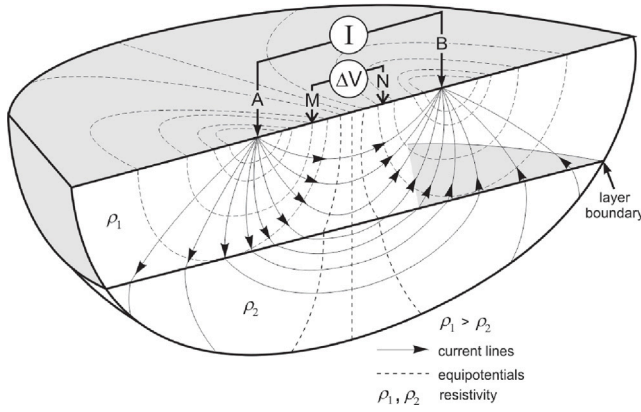


Fig. 5. Schematic representation of a resistance measurement in four-point arrangement over a layered half space (after Ref. 40).

4.1. Electric resistivity tomography (ERT)

ERT measurements provide robust results regarding the detection and in situ characterization of structural and sedimentary heterogeneities in claystone (Ref. 22, Ref. 39 and Ref. 33).

To determine the spatial resistivity distribution (or its reciprocal – conductivity) in the ground, a direct current (DC) is introduced through two point electrodes (A, B). The produced electrical field is measured using two other electrodes (M, N), as shown in Fig. 5. A point electrode introducing an electrical current I will generate a potential V_r at a distance r from the source. Assuming two current electrodes (A, B) that introduce a current I into the homogeneous half-space, the potential difference between the electrodes M and N is calculated as

$$\Delta V = \rho I \left[\frac{1}{2\pi} \left(\frac{1}{AM} - \frac{1}{AN} - \frac{1}{BM} + \frac{1}{BN} \right) \right], \quad (1)$$

where $\overline{P_1 P_2}$ denotes the distance between two points P_1 and P_2 . Replacing the factor in square brackets by C , we obtain the resistivity of the homogeneous half space as follows:

$$\rho = C \frac{\Delta V}{I} \quad (2)$$

The parameter C is called configuration factor or geometric factor. For inhomogeneous conditions it gives the resistivity of an equivalent homogeneous half-space. For this value the term apparent resistivity ρ_a in Ωm is introduced, which is normally assigned to the center of the electrode array. Multi-electrode resistivity meters enable the measurement of 2D resistivity surveys (2D imaging). The advantages of this kind of measurements are their high vertical and horizontal resolution along the profile.

The geoelectrical measurements were performed using the high-resolution earth resistivity meter 4-point light 10 W (Lippmann Geophysikalische Messgeräte), controlled by the software GeoTest (Geophysik–Dr. Rauen). Within the CD-A experiment, high-resolution borehole measurements along exploratory boreholes are used to characterize the rock. These are carried out once shortly after the drilling is completed using the BGR probe GB100-15 (Ref. 33). It consists of 100 electrodes with a spacing of 15 mm. The tool has a diameter of 80 mm (for typical hole diameters of 86 mm), but greater diameters can be adapted. The electrodes can be extended pneumatically for the required connection to the borehole wall. The spatial resolution is limited by 2 times the electrode distance (30 mm), the maximum depth of investigation is about 30 cm perpendicular to the borehole wall. The chosen electrode configuration represents a Wenner- α setup (Ref. 40).

A numerical inversion process is necessary for data analysis and interpretation. The measured apparent resistivities along the borehole are thus transformed into a spatial model, discretized into a distinct

number of elements of homogeneous resistivity, taking into account the electrode geometry. The forward operator in the inversion process is generally obtained by finite-difference (FD) or finite-element (FE) methods. For our data, the non-commercial software package BERT (Boundless Electrical Resistivity Tomography) developed by T. Günther (Leibniz Institute of Applied Geophysics, Hannover) and C. Rücker (BASE) is used^c. For further details, see Ref. 41 and Ref. 42.

4.2. Mini seismic method (MSM)

Seismic velocity measurements make use of elastic waves excited at defined source locations. The seismic wave field is recorded in terms of travel time and amplitude at certain receiver positions. Taking into account the acquisition geometry and travel time, propagation velocity of direct pressure (p-) waves v_p and secondary shear (s-) waves v_s can be calculated. Typical velocity values for the relevant lithology obtained by in-situ borehole measurements at the Mont Terri laboratory are discussed in Refs. 22, 43 and for laboratory investigations by Ref. 44.

The data indicates for the intact shaly facies of OPA P-wave velocity values around 2500 m/s, for the carbonate rich areas values between 3000 and 4000 m/s, and for the sandy facies values around 3000 to 3500 m/s. In fault zones an average decrease of 200 to 400 m/s relative to the base value was observed. Near excavations the measured values for the intact and damaged zones are varying, reflected by the highly variable seismic signal with fluctuations from 1500 m/s up to 2600 m/s as discussed by Ref. 22.

Depending on the respective survey geometry, travel time data can be used to deduce a velocity model along travel paths of the waves in two (2D) or three spatial dimensions (3D). If repeat survey data are available, temporal data can be included and velocity changes determined (4D), to permit geological process analyses. For the acquisition of interval velocity data, special interval velocity measurements (IVM) are carried out. In the CD-A experiment, a seismic borehole probe with one seismic source and seven receivers is moved stepwise along the surveyed boreholes with 5 cm intervals. At each step, the probe actuator and sensors are pneumatically attached to the borehole wall to establish sufficient mechanical coupling for the generation and detection of the emitted impulsive (“shot”) elastic waves (Fig. 6). The wave signals are recorded with seven receiver channels and one source channel (trigger channel).

4.3. Pulse tests and evapometer measurements

Information about the permeability of the host rock is gained with pulse tests in the boreholes BCD-A24 to BCD-A27 and with evapometer measurements in borehole BFM-D19. While the boreholes BCD-A24 and BCD-A26 are aligned in vertical direction, boreholes BCD-A25 and BCD-A27 are aligned in horizontal direction and BFM-D19 nearly in horizontal direction as presented in Fig. 3. The twin niches are located perpendicular to the strike of bedding with an inclination of 55–60° in the direction of the niches (145/59) which is summarized in Ref. 30 in detail. The pulse tests have been carried out in March 2021, 1.5 years after excavation of the niches, while the evapometer measurements have been carried out in summer (June/July) 2020 about 10 months after excavation.

In the pulse tests, a modified COMDRILL double piston packer system (<http://www.comdrill.de/>) with a static interval length of 10 cm is used. A non-reactive nitrogen gas at standard temperature and pressure was used as injection medium with an injection pressure up to 1 bar. The pulse test is characterized by the injection of the gas and hereafter a monitoring of the pressure evolution within the test interval. This allows the determination of the intrinsic permeability of a 3D volume in the near field around a 10 cm long borehole interval. A

^c <https://gitlab.com/resistivity-net/bert>

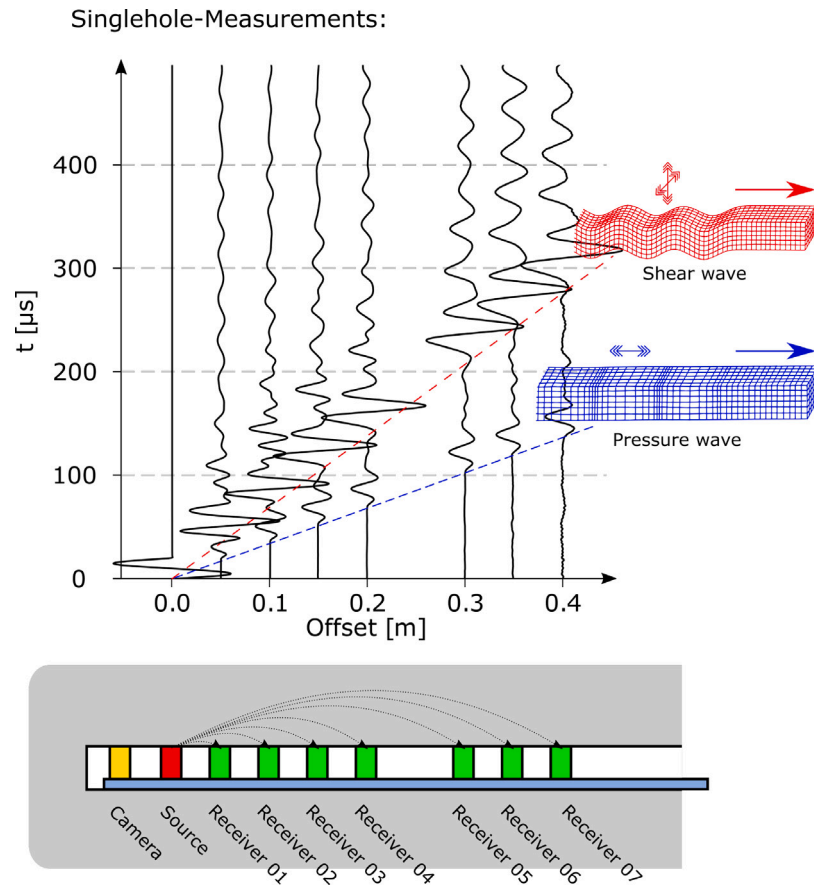


Fig. 6. Principle of seismic velocity measurements. Top: Emitted and recorded wave field at 0 to 40 cm distance. Blue: sketch of pressure wave propagation. Red: sketch of shear wave propagation. Below: geometry of interval velocity measurements (IVM, single hole). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

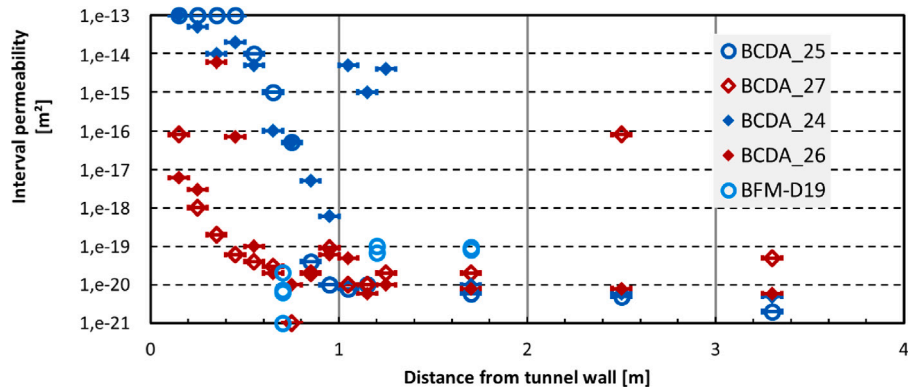


Fig. 7. Measured permeabilities with pulse tests (BCD-A24 to BCD-A27) and evapometer (BFM-D19) in open (blue) and closed (red) twin in horizontal (contour) and vertical (filled) drilling direction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

maximum error from experimental and evaluation sources is half the order of magnitude. More information about these tests is summarized in Ref. 22. The measured interval permeabilities in m^2 indicate that the permeability in the damaged zone can be up to seven orders of magnitude higher than in the undisturbed host rock as expected and also already described in Ref. 45 (two orders of magnitude higher) amongst others. These high permeabilities are found near the surface of the niche and decrease to the original value in the range up to 1.0 m distance from the surface. Furthermore, a significantly higher permeability within the first 1 m of the host rock around the open twin (presented in blue) than in the closed twin (presented in red) is found

in March 2021, 1.5 years after excavation (June/July 2020). In the vertical boreholes, this decrease is followed by another increase at depths around 1 m, followed by permeability values as in the undisturbed rock from a depth of about 1.2 m. As expected, the reported in-situ measured permeabilities are slightly lower than related results of laboratory tests. Ref. 46 reported higher hydraulic conductivities around $1 \cdot 10^{-13} \text{ m/s}$ computed from oedometric and constant-head permeability tests in the laboratory. Ref. 47 identified permeabilities for the intact host rock in a range between $6.9 \cdot 10^{-21}$ to $2.3 \cdot 10^{-19} \text{ m}^2$ for gas and water permeability measured in a permeameter called Hassler cell parallel and perpendicular to the bedding plane.

The evapometer measures the hydraulic conductivity K in m/s by measuring the humidity flux from a surrounding formation into a borehole at a 21 cm long test interval, the so-called matrix permeability. The intrinsic permeability k in m^2 as a parameter characterizing the host rock results from

$$k = K \cdot \mu / (\rho g) \quad (3)$$

with intrinsic permeability k in m^2 , hydraulic conductivity K in m/s, viscosity μ in Pa-s, density ρ in kg/m^3 and gravity g in m/s^2 . The measured permeability in m^2 is also presented in Fig. 7 indicating similar ranges than the pulse tests. Within the evapometer, a silica gel absorbs the air humidity which results in a change of the electrical field. For more information on the evapometer measurements the reader is referred to Ref. 16. As the evapometer measurements (carried out in summer 2020) are also influenced by the saturation of the host rock, the measurements indicate lower values for the humidity flux (up to two orders of magnitude) in the nearfield of the niche, characterized by a significant dehydration of the host rock. It has to be kept in mind that these assumptions are based on very few data points and hopefully can be convinced by further measurements in the future. The measured permeabilities are presented in comparison with other boreholes measurements in Figs. 8–11.

4.4. Time domain reflectometry (TDR/Taupe) measurements

The application of Time Domain Reflectometry, TDR, with the Taupe measurement system uses high-frequency voltage pulses on electrodes in contact with the material to determine its related apparent relative dielectric permittivity, ARDP. Due to the high ARDP of water (about 80 at room temperature) in relation to low ARDP of mineral materials (3...6), TDR method allows to identify water content in such environments. The traveling time of a pulse along the electrodes and back can be determined and the mean ARDP over the length of the electrodes calculated. Amplitude distribution of the TDR raw signal over the electrode length allows to determine the spatial resolution of ARDP values. A calibration function is necessary to calculate the water content from ARDP.

TDR tube sensors, consisting of two coupled parts of 2 m length each, are mounted in the vertical and horizontal boreholes in the twin niches. Because the sensors, mounted on the outside of the tubes, might not be in good contact to the host rock over its complete length, a certain time has to be maintained to establish reliable results for the ARDP determination. This is less significant for the horizontally installed tubes, but vertically installed sensors require up to 100 days until convergence of the host rock provides closing of possible air gaps around the tube sensors. Starting conditions and settling processes are not influenced by atmospheric conditions in open and closed niches, but dominantly by the geological situation inside the host rock, given by different layers crossing the bore holes.

The Taupe results presented in Fig. 10, Fig. 11, Fig. 8 and Fig. 9 show the timely resolved locally results for both niches. For the open twin, results of nearly horizontal borehole A3 (Fig. 8) show an increase in ARDP amplitude in a depth between 330 and 400 cm. Most significant, a rise in ARDP up to about 10.5 indicates accumulation of fluid at the back side of the borehole over time. Because of the inclination of the horizontal boreholes of ~ 10 degrees, fluid can progress by capillary rise or ingress the amount of fluid coming from faults in the host rock.

The ARDP distribution from sensors in vertical borehole A4 (Fig. 9) remains nearly constant with only small variations in depth and a total range of ARDP between about 7 and 8. Slightly higher values are concentrated in depth between 60 cm and 250 cm.

In the closed twin the nearly horizontal borehole A7 (Fig. 10) shows a configuration that is nearly stable after about 110 days indicating a zone of slightly higher ARDP centered at about 90 cm and ranging from about 40 cm to about 200 cm. ARDP range is between 7.5 and 8.5 with

only low indication of additional fluid access, but on settling processes in the host rock around the bore hole.

The rear part of the coupled TDR tube sensor in vertical borehole A8 (Fig. 11), closed twin, was not completely inserted into the bore hole and the sensor start is about 15 cm out of the bore hole mouth. Results show an ARDP distribution which varies only slightly with ARDP values ranging from 7 to 9 in a depth range between 100 and 250 cm. Additionally, a zone of ARDP values around 8.5 exist around depth 285 cm and at the backside of the sensor up to ARDP values of 9. Because of its stability, this might show structural properties of the host rock.

The structure of the vertical boreholes, A4 and A8, is comparable, but absolute values are higher in A8. Responsible for the difference in the vertical boreholes could be the different geological situation at the location of the boreholes. As expected, the nearly horizontal boreholes A3 and A7 show different structures and different ARDP value ranges due to the distance between them. Despite the fact of strong geology changes and different measuring orientations of the TDR sensors, ARDP values of all measurements vary only in a narrow range. Data of borehole A3 indicate a restricted inflow of fluid at the sensor end, which is reducing over time.

At the front part of the TDR sensors ranges of lower ARDP exist, possibly indicating the existence of an EDZ.

4.5. Nuclear magnetic resonance measurements (NMR)

Geophysical applications of nuclear magnetic resonance (NMR) estimate the water content of porous geo-materials by detecting the response signal of ^1H nuclei after electromagnetic stimulation. The application of an electromagnetic excitation forces the spins of the proton nuclei in the water molecules to tilt away from their equilibrium. After shutting off the excitation field, the spins relax back to equilibrium in a static magnetic field, which is captured by the NMR sensor as an exponentially decaying signal. The more ^1H protons are included in this process, i.e., the more water molecules are inside the sensitive volume of the measurement, the stronger is the detected signal. A simple linear relationship transfers the measured signal amplitudes into estimates of water content. We determine the signal amplitude in this study by mono-exponential fitting of the measured NMR data. Besides water content, also pore space information can be provided from the relaxation behavior of NMR signals such as pore size distribution, capillary pressure curve, and permeability.^{48–50} However, the interpretation of relaxation times for pure clay rock is an issue of recent research^{50–52} and is not part of the CD-A project at this early state.

The drilling program in 2021 was accompanied by measurements using mobile NMR technologies for investigating the drilled rock cores directly after taking them, the so-called single-sided NMR⁵³ and a mobile core scanner.⁵⁴ Interpreting the results, one important difference has to be kept in mind: While the single-sided NMR device measures a thin slice with a thickness about 200 μm , the mobile NMR scanner interprets several hundreds of cubic centimeters. Using a mobile NMR core scanner provides the opportunity of keeping the core material undamaged for subsequent studies. The working principle is the same as for conventional laboratory NMR devices⁴⁹ with the limitation that the mobility, i.e., flexible installation and usage anywhere in the tunnel, comes at the costs of a light-weighted and thus weaker permanent magnet inside. This leads to a worse signal-to-noise ratio compared to the data quality of usual laboratory NMR data. Immediately after taking the drilling cores, the core material was covered with plastic foil and step-wise pushed through the device. Measurement points were determined with a spatial resolution of about 20 cm, which corresponds to the length of the sensitive range inside the Scanner. In doing so, a more or less continuous coverage of the cores could be realized. The results are depicted in Figs. 8–11 and compared to water contents as measured in the laboratory by weighing at 105 °C and using the CCM. The results are in agreement. In contrast to the drilling program in

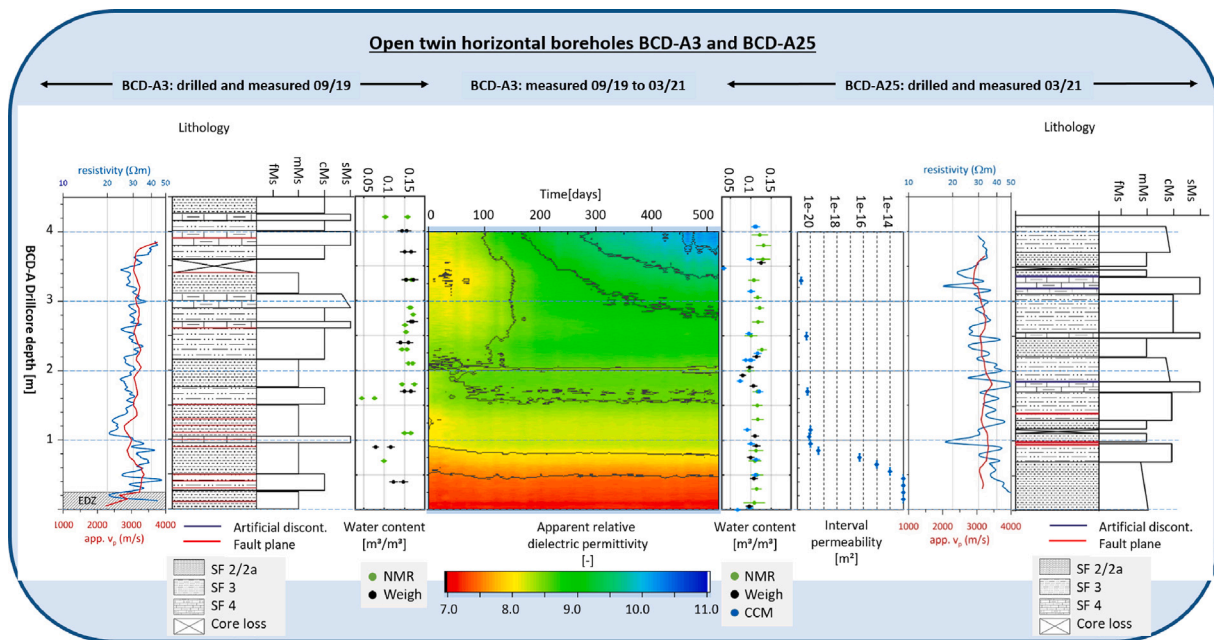


Fig. 8. Comparative presentation of the lithologic interpretation, the measured electric resistivity in Ω , seismic velocity in km/s, apparent relative dielectric permittivity, and permeability in m^2 as interpreted from the horizontal boreholes and related cores taken in September 2019 and March 2021 in the open twin niche.

2021, the core water contents during the initial core drilling in 2019 had to be measured using single-sided NMR, for which specific samples had to be prepared out of the cores. These results can also be found in Figs. 8–11.

4.6. Calcium carbide method (CCM)

The Calcium Carbide Method (CCM) is a technique to measure the water content of a soil sample of a few cubic centimeters based on this chemical reaction:



By measuring the excess pressure due to the formation of the acetylene gas (C_2H_2) (and ethin C_2H_2) in a closed reactor, one can decipher the amount of water (H_2O), which reacted with the Calcium Carbide (CaC_2). This technique is well adapted to measurements in-situ. In the CD-A experiment, it has been first used in March 2021. For this purpose we were using the “Carbide Moisture Meter” equipment from Radtke Messtechnik. As soon as a core was drilled, we took a 20.0 g sample of the soil and ground it in the calibrated pressure bottle using steel balls and a stirring rod. Then an ampule with 7.0 g (± 140 mg) Calcium Carbide was introduced and the container was sealed. By shaking the calibrated pressure bottle, the ampule cracks and the CaC reacts with the free water molecules. After ten minutes of reaction, we measured the increase of pressure on the manometer of the sealed bottle and determined the water content in the sample.

During the drilling program in 2021, the CCM was applied in addition to the NMR water content acquisition (see Figs. 8–11).

5. Comparative presentation and interpretation

A comparative representation of the geophysical measurements and geologic interpretation of the boreholes in horizontal and vertical direction is presented in Figs. 8–11 for both niches.

To quantify the intra-facies variability of the Opalinus Clay at the Mont Terri Rock Laboratory, the main facies types are further subdivided into several subfacies (SF) types. Each subfacies type is characterized by a distinctive color, lithology, grain size, fabric and increasing carbonate content (Ref. 34):

- SF2: Medium-dark gray, sparsely bioturbated, continuous to discontinuous planar-parallel-laminated, argillaceous-siliceous medium to coarse-grained mudstone including sparse bioclastic fragments.
- SF2a: Dark gray, argillaceous-siliceous medium-grained mudstone, with less sparsely to moderately cemented silt-rich laminae in comparison to SF2.
- SF3: Medium gray, moderately bioturbated, lenticular-laminated or interwoven, siliceous-argillaceous sandy mudstone with common bioclastic fragments.
- SF4: Medium-light gray, strongly bioturbated, flaser-bedded to homogeneous siliceous-calcareous sandy mudstone with abundant bioclastic fragments.

The lithologic characterization indicates various subfacies types are occurring along the boreholes as can be seen in all drilling profiles. Please note that the figures show a generalized overview, further variations can occur at the cm- and sub-cm-scale. The Opalinus Clay in the vicinity of the niches is characterized by the subfacies types SF2 to SF4 according to the classification of Ref. 34. Subfacies type SF1 (homogeneous, argillaceous fine to medium grained mudstone) and SF5 (Light gray, medium to coarse grained calcareous-sandy mudstone to limestone) were not encountered in the upper sandy facies of the CD-A drill cores.

Regarding Figs. 8–11 it becomes apparent, that a rather high-lithological intra-facies variability subperpendicular to bedding is widespread, as indicated by various small-scale subfacies variations in this particular orientation. On the other hand, the data indicates a rather low lateral variability parallel to bedding.

The lithologic profiles correlate to the measurements of the electric resistivity. Varying resistivity in high-resolution ERT measurements can be attributed to differences in clay mineral and carbonate content or structural features (see below). Especially the vertical boreholes indicate characteristic sequences with different resistivities with values between 10 to 50 Ωm which correspond to varying subfacies types. In the horizontal boreholes, this correlation is not as pronounced, since in the boreholes parallel to bedding the ERT results depend on the orientation of the high-resistivity carbonate layers as well as on the measurement geometry.

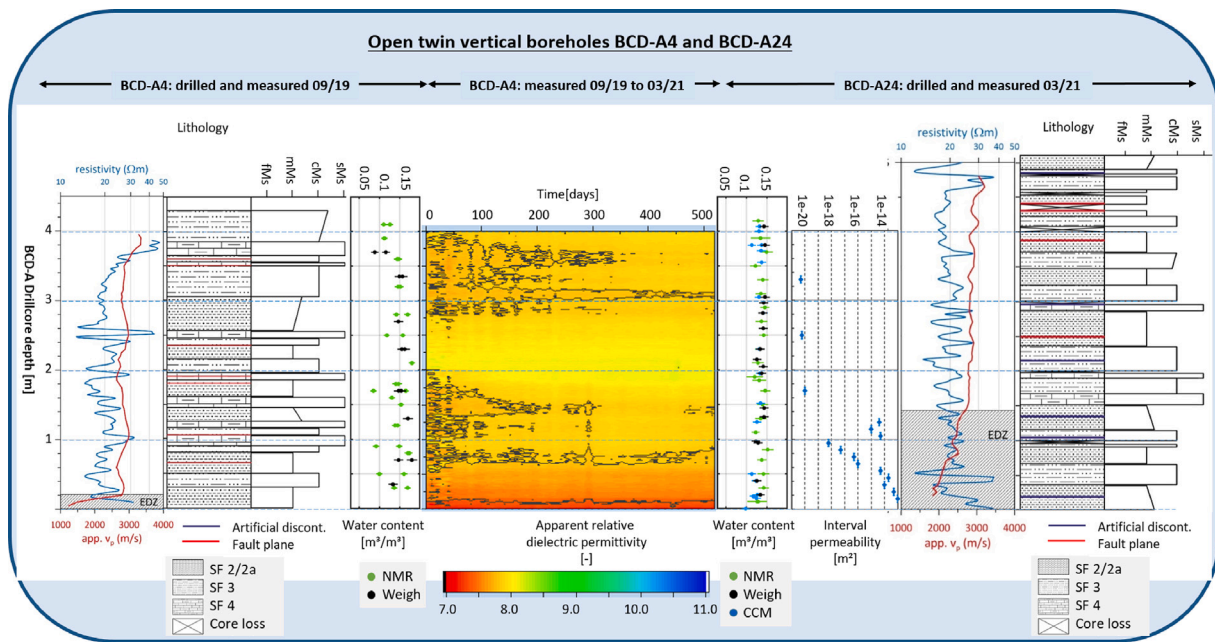


Fig. 9. Comparative presentation of the lithologic interpretation, the measured electric resistivity in Ω , seismic velocity in km/s, apparent relative dielectric permittivity, and permeability in m^2 as interpreted from the vertical boreholes and related cores taken in September 2019 and March 2021 in the open twin niche.

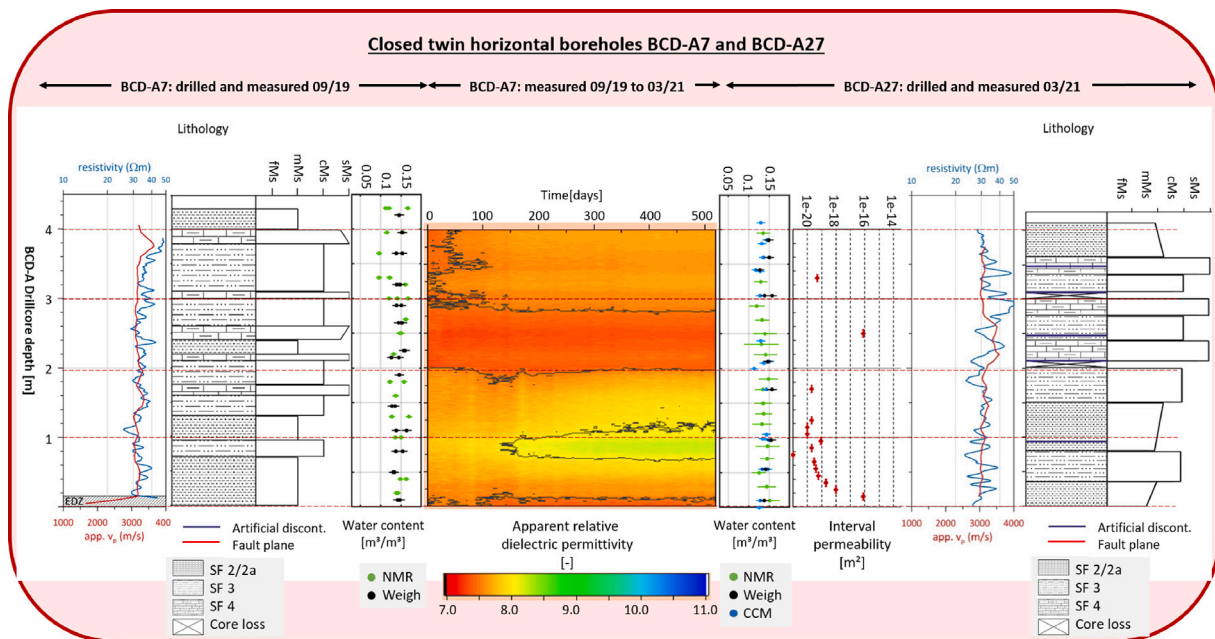


Fig. 10. Comparative presentation of the lithologic interpretation, the measured electric resistivity in Ω , seismic velocity in km/s, apparent relative dielectric permittivity, and permeability in m^2 as interpreted from the horizontal boreholes and related cores taken in September 2019 and March 2021 in the closed twin niche.

Generally, the intervals with higher carbonate content (subfacies types SF3 and SF4) are characterized by increased values of electric resistivity $> 30 \Omega m$. On the other hand, the measurements that focus on the water content like Taupe, NMR, and CCM measurements show no clear correlation of the water content with the subfacies type in question. In addition, the measurements of the interval permeability derived from hydraulic pulse test indicates no clear correlation to the lithologic profile. Nevertheless, due to general fluctuations in the measured values as well as the rather low number of measurements to date, no clear statement regarding the correlation between subfacies type/lithology and measured water content is possible so far. A comparison of different facies types can be found for example in Ref. 33

while the measurements in the context of the CD-A experiment focus on different subfacies types. The small-scaled water content measurements (CCM and NMR) do not show the expected increasing trend with increasing depth and differ to the Taupe results in this regard. The reason for this is most likely the different integration ranges of these measurements. The measured permeabilities show the expected increase near the excavation and also some measurement anomalies, such as the elevated permeability ($10^{-16} m^2$) at 2.5 m borehole depth in borehole BCD-A27, which can be well explained by the presence of an artificial discontinuity as it is reported in the lithology plot of 10 by the blue line and also in the borehole scan, showing a distinct

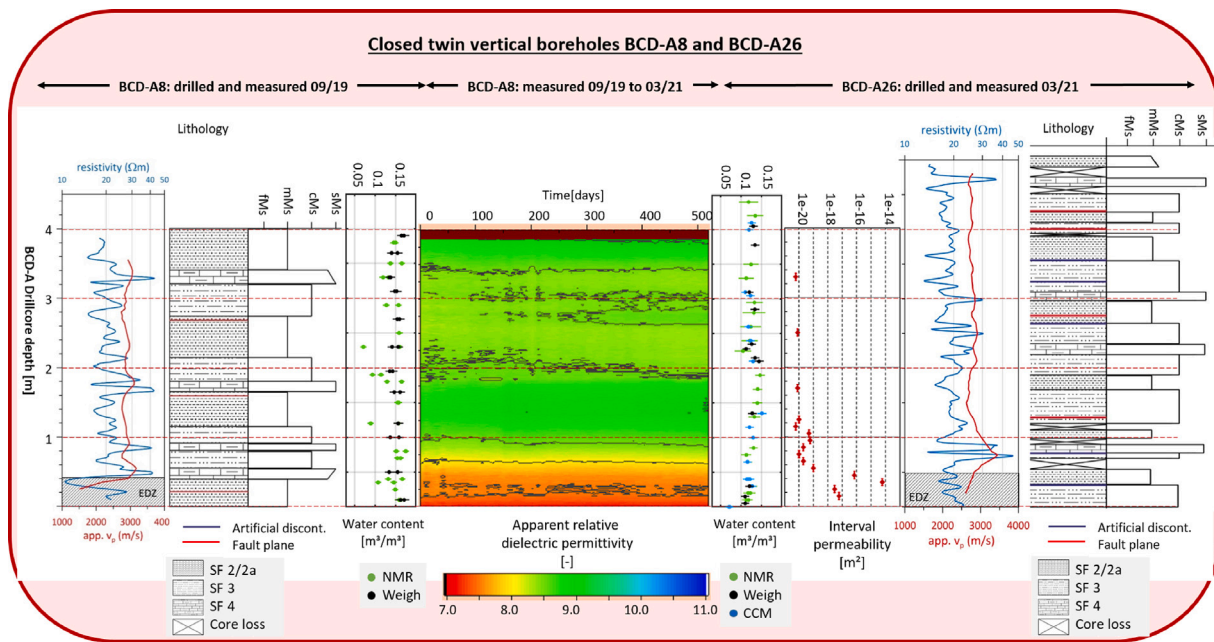


Fig. 11. Comparative presentation of the lithologic interpretation, the measured electric resistivity in Ω , seismic velocity in km/s, apparent relative dielectric permittivity, and permeability in m^2 as interpreted from the vertical boreholes and related cores taken in September 2019 and March 2021 in the closed twin niche.

horizontal fracture structure at 2.4 m (given in the supplementary data for borehole BCD-A27).

In addition to the lithological variability, geophysical measurement repetitions can provide information about the zone, which is affected by the excavation: What is their extent and what is the degree of influence? The geophysical measurements presented here are to be seen as a time-related statement - they do not provide information about the temporal evolution and a possible associated reversibility. However, the geophysical monitoring is an important part of the long-term program of the CD-A project and might provide insights also on EDZ dynamics in the future. The evolution of the EDZ has already been in the focus of previous investigations as for example discussed by Ref. 55 providing information about the hydro-mechanical evolution of the EDZ as transport path for radionuclides and gas. Especially the MSM indicates the extent of the EDZ very clearly as presented in Figs. 8–11 and indicates an extent of the EDZ varying between 20 cm and 50 cm, measured immediately after the excavation of the niches (BCD-A3, BCD-A4, BCD-A7 and BCD-A8). The EDZ is characterized by clearly reduced p-wave velocities varying between 1000 m/s and 2600 m/s, compared to velocities between 2800 m/s and 3500 m/s for the intact rock. 1.5 years later (in the boreholes BCD-A24 to BCD-A27), the EDZ can only be measured clearly in the vertical direction of each twin, while there is no clear change in the measured velocities in the horizontal directions. While the open twin is characterized by a significant increase of the measured EDZ, there is only a minor increase in the closed twin. As resistivity is dependent on porosity and water saturation (Ref. 56), some features of the unsaturated EDZ are also visible in the geoelectric profiles, showing increased resistivities $\rho > 25 \Omega \text{ m}$. This effect can be found in all resistivity measurements of the open twin and also in the early horizontal measurement (BCD-A7) of the closed twin.

The measurements of the apparent relative dielectric permittivity with the Taupe sensor as well as the permeability measurements by the pulse tests provide information about the characteristics in the nearfield of the niches. The measurements of the interval permeability in boreholes BCD-A24 to BCD-A27 have been carried out 1.5 years after the excavation of the niches (Fig. 7). The extend of the zone showing a significant increase of the permeability is significantly higher in the open (up to 1 m) than in the closed niche (up to 50 cm), with max.

permeability values varying from usually (one outlier is measured at a depth of about 40 cm) 10^{-16} m^2 (closed) to 10^{-13} m^2 (open). The contrasting climatic conditions in the open and closed niche might be a reason for the different extent of the EDZ and the differences of several orders of magnitude in interval transmissibility.

The measurements of the apparent relative dielectric permittivity with the Taupe sensors indicate varying values in the nearfield of the niches compared to the deeper parts of the boreholes. Due to the continuous evolution of the ARDP, the range of the EDZ is not easy to distinguish in the dataset, but comparing TDR results in this area with ERT results indicate ARDP values below about 7.5 as a delimitation of a zone away from the niche surface (Open twin about 40 cm for A3 and A4, closed twin: about 15 cm for A7 and about 25 cm for A8). Focusing on the temporal evolution of the nearfield effects in Figs. 8–11, the measurements indicate nearly constant values for both niches and show no significant apparent increase of the damaged area around the niches. Considering all three methods, the extent of the affected zone in the nearfield of the niches is in the same range, nevertheless, the exact values may vary e.g. in the horizontal borehole of the open twin.

As several fault zones and discontinuities are crossing the studied section of the Opalinus Clay, the interpretation of the measured values with regard to existing heterogeneities is of interest. The comprehensive geologic interpretation provides information about artificial discontinuities as well as the information regarding core losses, the location of different subfacies types, the orientation of the bedding planes and the lithologic characteristic of the host rock. Taking this data into account, it is possible to relate some of the observed anomalies in permeability (outside the EDZ) to discrete artificial and tectonic features (see Fig. 10 where these effects can be correlated with the diagram). For example, the elevated permeability ($10^{-17} \text{ m}^2 \cdot \text{m}$) at 2.5 m borehole depth in borehole BCD-A27 can be well explained by the presence of an artificial discontinuity (blue line in the lithology plot in Fig. 10).

Discontinuities may vary significantly with regard to their mechanical and hydraulic properties and the interpretation of the relevance of fault zones concerning their impact on coupled thermal-hydraulic-chemical effects is challenging. Fig. 12 provides an overview of the observed faults and fault zones in the vicinity of the

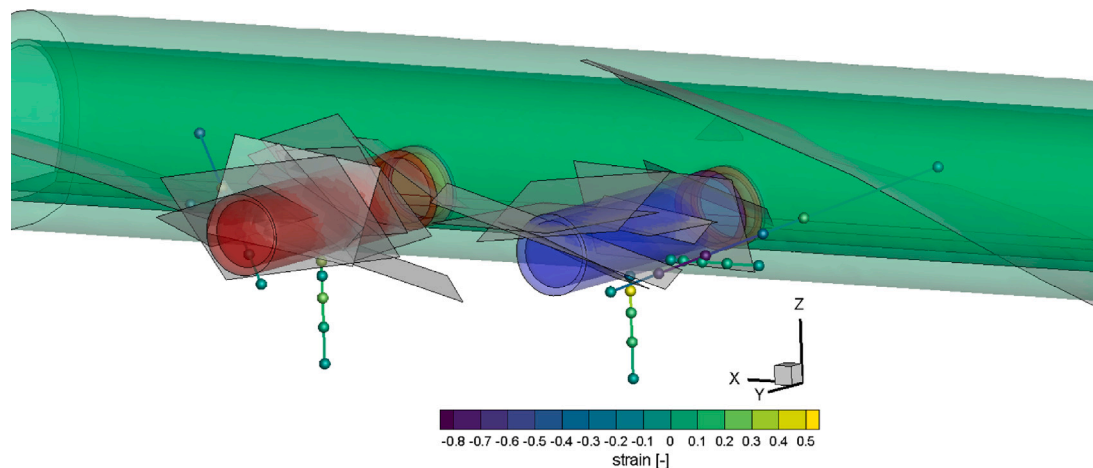


Fig. 12. Observed fault zones in the vicinity of the twin niches and extensometer measurements of strains.
Source: Taken from Ref. 32.

twin niches. For more information about the structural interpretation and the measured displacements, the reader is referred to Ref. 32. Comparing the lithologic description and the geophysical/geohydraulic data with the occurrence of fault zones results in some interesting observations which are summarized in the following section.

6. Results and discussion

The presented methods of investigation aim on various parameters and build a useful combination to investigate the characteristic of the claystone at a meter-scale around the niches. The subfacies types can be identified with electric resistivity measurements and lithologic description. Given that the measurements of lithology, water content and permittivity exhibit no clear correlation, it becomes evident that the primary objective of a numerical model is to represent the overarching coupled behavior rather than individual subfacies. Consequently, a homogeneous modeling approach might be sufficient to capture this behavior in the presence of diverse subfacies types of Opalinus Clay. However, there has to be a particular focus on the EDZ around the excavations as this area is characterized by various heterogeneities and an ongoing evolution over the years. Nevertheless, also the presented measurement examinations provide information about the extent of the EDZ around the niches (ERT, MSM, Taupe, Pulse test) at the borehole locations, their evolution 1.5 years after excavation and the impact of the climatic conditions on the EDZ. While the initial extent in both niches and both directions results to 20 to 50 cm, 1.5 years later we found no clear indications of the EDZ (by ERT measurements) in horizontal directions of both niches, while the EDZ in vertical directions results to 140 cm in the open and 50 cm in the closed twin, indicating a significant impact of the climatic conditions on the EDZ evolution. This impact is also seen in the permeability measurements in horizontal as well as vertical direction - permeability measurements 1.5 years after excavation with hydraulic pulse tests display different behavior in the closed (affected area up to 50 cm) in comparison to the open twin (affected area up to 100 cm) and differences of several orders of magnitude in interval permeability. The most obvious reason for this result - which has to be confirmed by further measurements - seems to be the different climatic conditions of the niches. Nevertheless, aspects such as the characteristics of the claystone, the orientation in relation to stress field and bedding and the climatic boundary conditions must be taken into account. Nevertheless, the Taupe measurements indicate no remarkable change of measured permittivity in the EDZ over time.

The Opalinus Clay in the vicinity of the twin niches is characterized by several discontinuities like fractures and fault zones with varying impact on the hydraulic as well as mechanical conditions. Heterogeneities

such as faulted areas as well as micro cracks can be investigated due to the lithological description of boreholes, geologic characterization of the host rock, the electric resistivity measurements (especially seasonally affected fractures), and potentially the Taupe measurements, indicating an impact of fractures on the hydraulic effects. While there seems to be a correlation between some of the observed artificial discontinuities and the electric resistivity as well as the permeability, the overall impact on the water content seems to be low in the range of the borehole length of 4 m. Nevertheless, the impact of these kind of heterogeneities varies significantly from case to case and underlines the very complex behavior of claystone. The resulting change of material parameters may be of significant importance concerning a potential safety case of a long-term HLW repository and therefore is planned to be in the focus of further investigations, aiming particularly on the decimeter-scale around the excavations.

The water content is a key parameter for the coupled effects in the vicinity of the twin niches, which is observed in this project by a combination of various methods. In Ref. 31, water content measurements performed along the surface of the niches were correlated with electrical resistivity data and numerical modeling. Seasonal effects have been described and the statistical evaluation of such temporal data has been argued and correlated with geological characterization. The investigations enable a temporal description of the water content evolution around the niches, indicating the highly heterogeneous near field effects. Here, NMR and CCM measurements have been successfully conducted using the fresh core material immediately after drilling to provide water content profiles as original as possible. The two methods yielded very similar results that could be validated by subsequent weighing in the laboratory. These water content profiles with values scattering within the range of 10 to 16 vol% do not show clear trends or systematic variations and thus support the idea of a homogeneous water content model for the numerical HM simulations at the average of the observed values. However, further investigations are necessary to access the relationship between the lithological subfacies types and corresponding water content. A clear relationship does not seem to exist regarding our data base to date, which is not intuitive. One challenge in this context is the different dimensions at which the individual methods monitor the water content and their individual measurement principles ranging from a few μm (single-sided NMR), over a few cubic centimeters (CCM) up to several hundreds of cubic centimeters (mobile NMR). Further laboratory research, which considers these methodological issues, is underway to characterize how the pore water in the Opalinus Clay reacts at the μm to the cm scale on atmospherical changes.

Although indicated by Taupe, an increasing water content with increasing depth in the range of the EDZ was not observed with NMR

and CCM. We address this observation to the fact that these methods work at a smaller scale than Taupe and are thus not able to probe volumes large enough to cover the corresponding changes, i.e., most likely cracks and fissures.

The long-term CD-A experiment is a favorable environment to further evaluate the presented investigation procedure aiming on a quantitative evaluation and verification of the used measuring and modeling approaches. Additional investigations like laser scans, convergence measurements, and extensometer measurements are carried out within and around the niches. They focus on the deformation behavior of the host rock around the niches and are discussed in combination with geologic interpretations, numerical modeling approaches and visualization techniques in Ref. 57, and Ref. 32. Investigations focusing on the pressure evolution conducted by mini-piezometer, psychrometer and thermo-hygrometer measurements in combination with numerical modeling procedures are underway. Within a HM coupled framework, the numerical procedure should describe the changes of water content as observed during the experiment. Furthermore, it is important to study the excavation-induced effects, e.g. size of EDZ, accounting for the mechanical anisotropic behavior of the rock that affects the EDZ shape/formation and, consequently the pore pressure distribution. Note that the initial in-situ stress state as well as initial pore pressures also have an influence on how the excavation will deform and how the water content will distribute.⁵⁸ Furthermore, the ability of the numerical model to reproduce such behavior must be critically evaluated.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ijrmms.2023.105624>.

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